

Challenge – Visualisation d’une base de données de films

Methodology

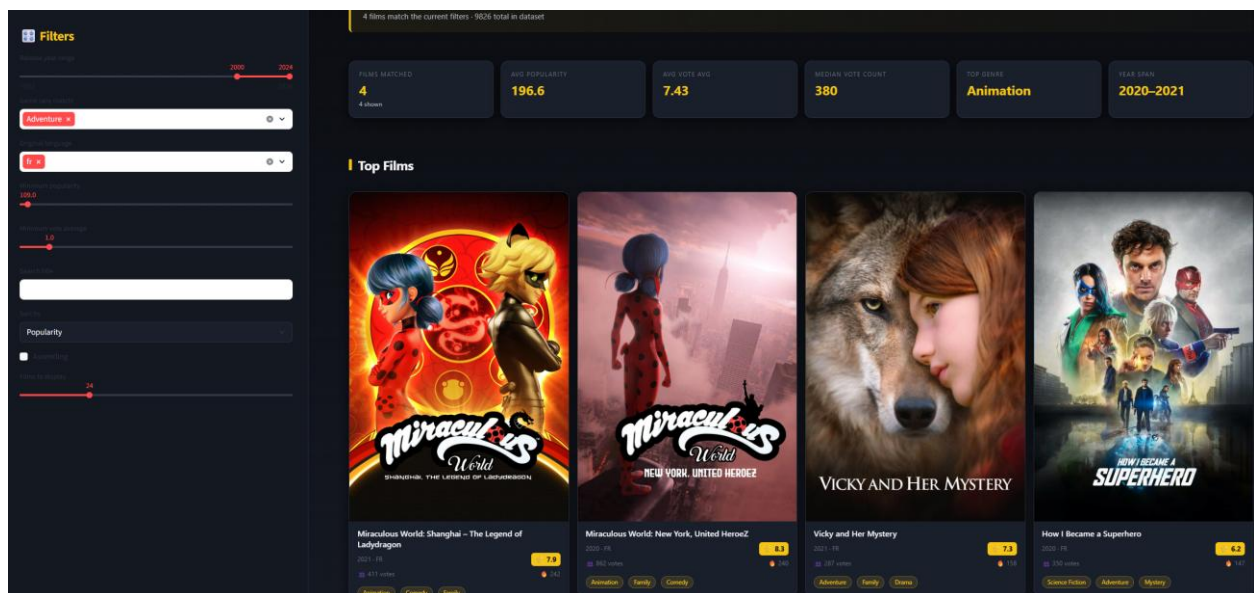
The project began by reading the statement of purpose (SOP) and downloading the CSV dataset. Jupyter Lab was used with standard libraries — Pandas, Matplotlib, Seaborn, SciPy, and Statsmodels — for data loading, type conversion, cleaning, and exploratory data analysis (EDA). Missing values (NaN, NaT) were identified and excluded.

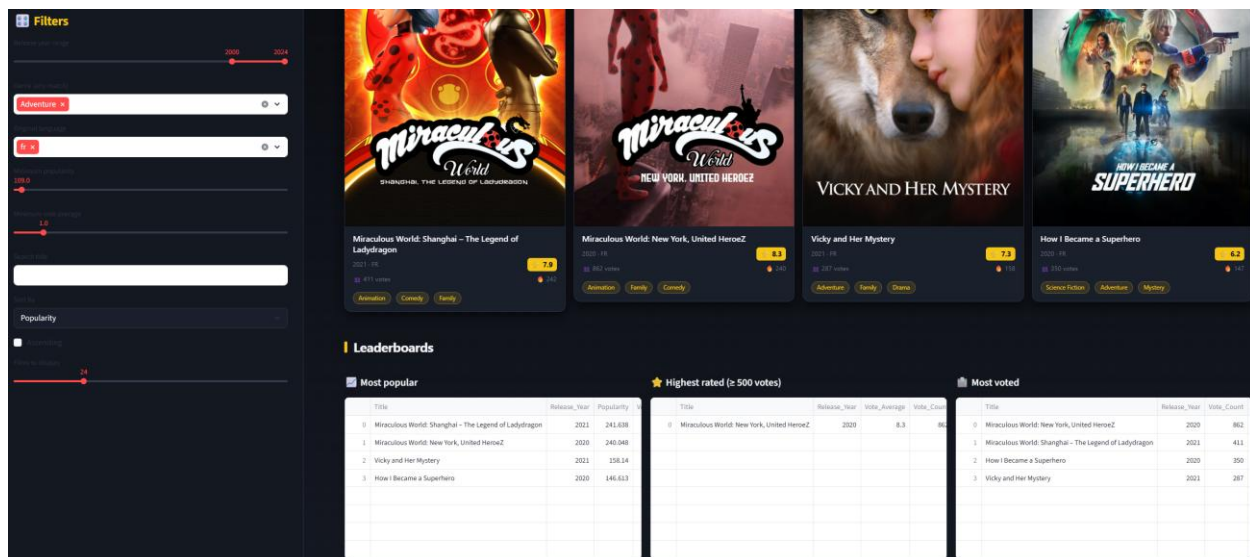
Note: The ‘Title’ field remains suspect — an OCR/multimodal model (e.g. glm-ocr) could validate it against the CSV. Missing ‘Overview’ values could be recovered via LLM API calls combined with web search.

Poster images were downloaded asynchronously using httpx with asyncio (more efficient than the requests library). After completing the EDA, Streamlit was chosen for the UI over Gradio, since no ML model was deployed (Gradio better suits displaying model performance metrics). Initial Gradio code produced poor rendering, so Alibaba’s Qwen was consulted to refine the Streamlit UI layout, using the Jupyter notebook as a reference.

Main Functionalities

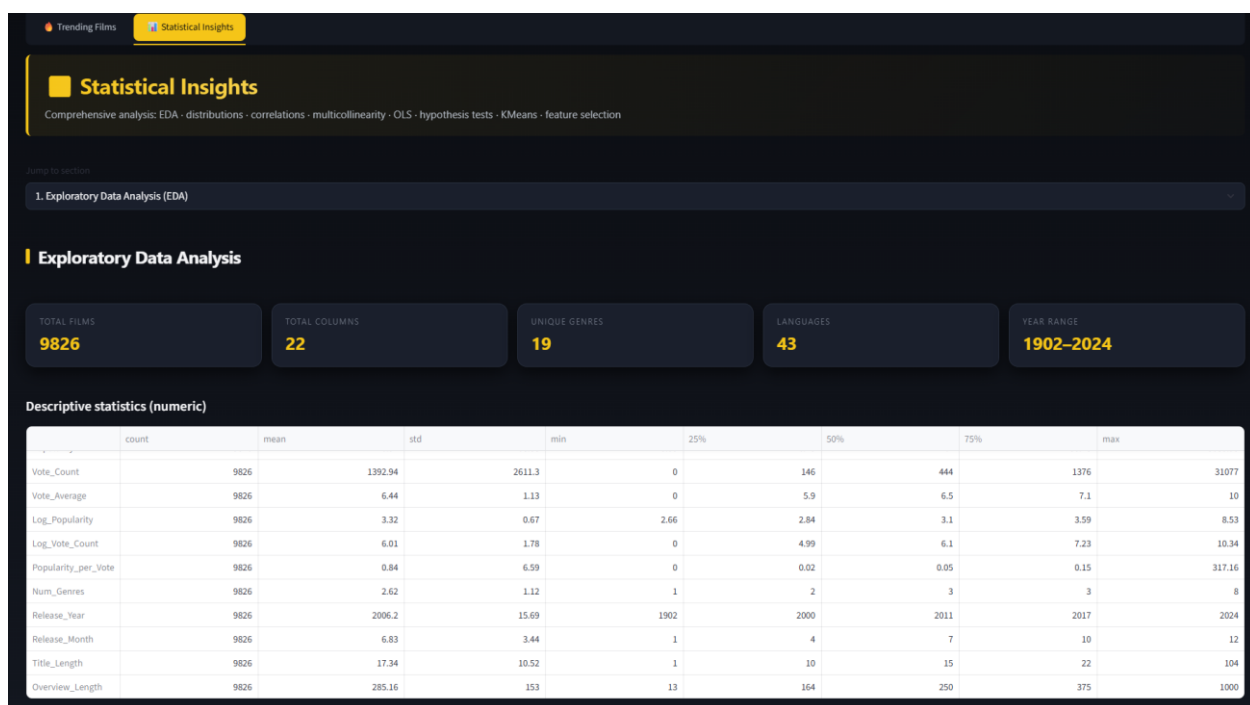
The UI displays movies with sorting options: Popularity (default), Vote Average, Vote Count, and Release Date. Additional filters allow narrowing by year, genre, popularity threshold, and result count. A leaderboard tab enables quick top-movie selection, and a Statistical Insights tab provides deeper analysis of the dataset. Only cleaned data is shown in the UI; records with incoherent values are examined in the Jupyter notebook.





Reproducibility of the Code

A README.md has been provided, detailing environment setup, how to run the code, and how to access the UI. The exploratory Jupyter notebook is also included for review. The full codebase is available at: <https://github.com/agarwalamit081/movie-analysis>.



Limitations and Possible Improvements

The SOP and scope of work (SOW) need more precise definition; depending on business requirements, classification or regression models could be developed. Column definitions, sampling methodology, and class balance information are absent — the data shows skewness (more films post-2000; Vote_Count and Popularity are skewed), which limits direct use in ML models. AI-assisted imputation could automate recovery of missing data. Finally, no containerization (Docker) was applied, as the project involved only a small number of scripts. Text2sql methods and voice assisted querying can be used in conjunction with multimodal LLMs to make the application more interactive.